Market Segmentation Analysis via K-Means Clustering

A Data-Driven Approach to Retail Customer Profiling

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Abstract

Customer segmentation is an essential analytical process for modern retail environments seeking to optimize marketing efficiency and resource allocation. This project employs unsupervised machine learning, specifically K-Means clustering, to autonomously partition a dataset of 200 mall customers into distinct behavioural profiles based on demographic and financial features. Through comprehensive Exploratory Data Analysis (EDA) and rigorous statistical cluster validation using both the Elbow Method and Average Silhouette analysis, we identified six predominant customer segments. These segments span from High Income, High Spending (Prime Target Customers) to Low Income, Low Spending (Sensible Spenders), with intermediate profiles that provide actionable marketing insights. To operationalize these findings, an interactive Graphical User Interface (GUI) was developed using Python's tkinter library, allowing non-technical stakeholders to perform instantaneous predictive profiling of new consumers. The project demonstrates how unsupervised machine learning can transform raw transactional data into strategic business intelligence, enabling retail teams to execute precision-targeted marketing initiatives.

Keywords: K-Means clustering; customer segmentation; retail analytics; unsupervised learning; Elbow Method; Silhouette analysis; GUI application

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1. Problem Statement

Understanding consumer behaviour is a pivotal challenge in the retail industry. In modern retail environments, marketing teams face the persistent problem of campaign inefficiency and suboptimal resource allocation. Without knowledge of customer characteristics, purchasing patterns, and behavioural profiles, marketing initiatives remain generalized and undifferentiated, resulting in budget wastage and poor conversion rates.

The primary objective of this project is to analyse a dataset of mall customers to partition them into mutually exclusive, homogeneous segments. By identifying these distinct target clusters, marketing teams can formulate precise, data-driven strategies tailored to specific demographics and behavioural patterns. This segmentation enables the retail business to maximise both profitability and customer satisfaction through personalised engagement strategies.

The analysis addresses the following key questions: How many distinct customer segments exist within the population? What are the defining characteristics of each segment? How can stakeholders operationalize these insights to make real-time predictions for new customers?

2. Dataset Description

The empirical data utilized in this study was sourced from Kaggle, a publicly available machine learning repository. The Mall Customers Dataset comprises 200 customer records, each containing the following features:

Feature	**Data Type**	**Description**

CustomerID	Numerical	Unique identifier for each customer
Gender	Categorical	Male or Female classification
Age	Numerical	Customer age in years (range: 18-70)
Annual Income	Numerical	Yearly income in thousands of dollars (k$)
Spending Score	Numerical	Proprietary metric (1-100) based on purchasing behaviour

The Spending Score is a proprietary metric assigned by the retail mall based on customer purchasing behaviour, frequency, and monetary expenditure history. This metric encapsulates the mall's assessment of each customer's value proposition and purchasing propensity. The dataset contains no missing values and exhibits balanced representation across demographic categories.

3. Exploratory Data Analysis (EDA)

Prior to machine learning model implementation, rigorous Exploratory Data Analysis was conducted to understand the statistical distributions and characteristics of the dataset. This preliminary phase is essential for identifying patterns, anomalies, and feature interactions that inform subsequent analytical decisions.

3.1 Gender Distribution

The dataset exhibits a minor skew towards female customers, with 56% female and 44% male representation. This slight demographic imbalance does not pose methodological concerns for clustering analysis, as K-Means operates independently of class frequency. Both barplot and pie chart visualisations were generated to communicate the count and proportional representations to stakeholders. Figure 1 illustrates this gender distribution, with the left panel showing absolute counts and the right panel displaying proportional representation.

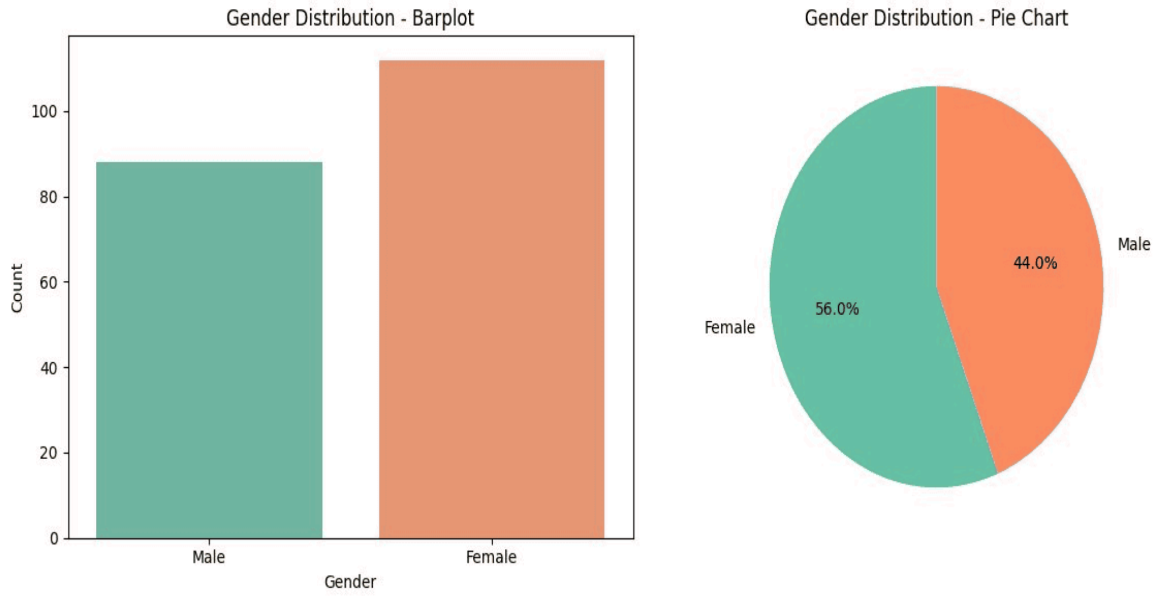

Figure 1: Gender distribution of the customer dataset. The barplot (left) displays absolute customer counts by gender, whilst the pie chart (right) shows proportional representation (56% female, 44% male).

3.2 Age Demographics

Age distribution analysis reveals a higher concentration of customers in the late twenties to early thirties cohort. A histogram with Kernel Density Estimate (KDE) overlay was employed to visualize frequency distributions, whilst a boxplot was generated to identify quartiles and detect potential outliers. The median age is approximately 36 years, with the interquartile range spanning from 28 to 49 years. This age concentration suggests that the retail establishment attracts primarily young professionals and early-career customers. Figure 2 displays both the histogram with KDE overlay and the boxplot summary statistics.

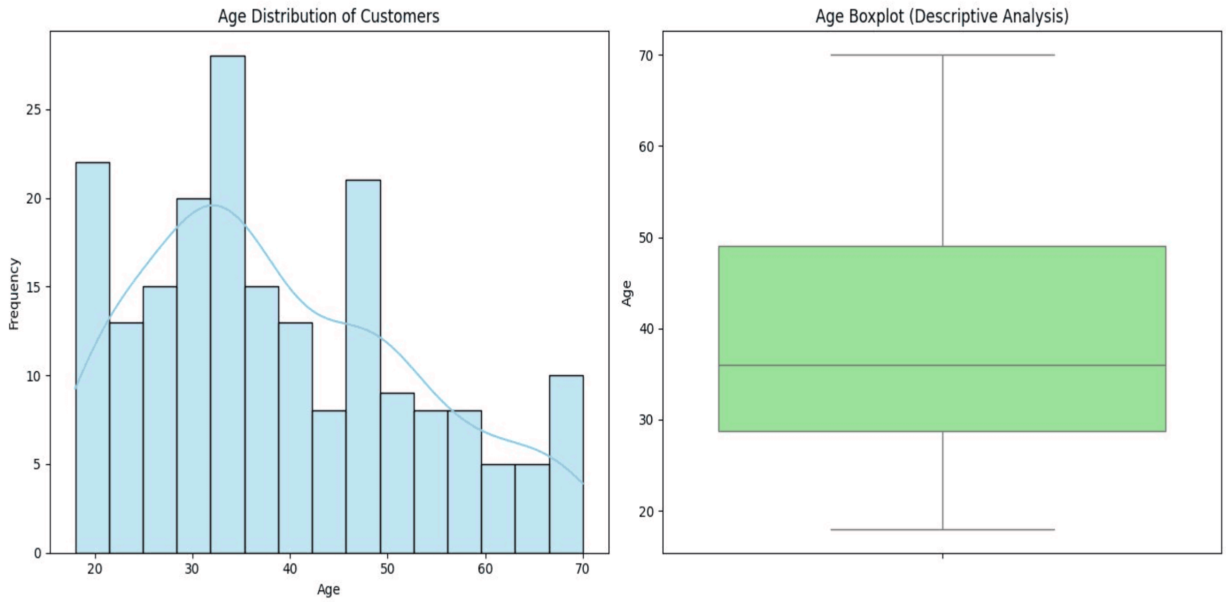

Figure 2: Age demographics of the customer base. The histogram with KDE overlay (left) illustrates frequency distribution across age bands, whilst the boxplot (right) identifies quartiles, median (Q2 ≈ 36 years), and outliers.

3.3 Financial Metrics

Annual income follows an approximately normal distribution centred around the 50,000 to 75,000 dollars range. The density plot indicates minimal skewness and approximately 68% of the customer base earns between 40,000 and 90,000 dollars annually. This distribution suggests a predominantly middle to upper-middle income clientele. The spending score distribution shows greater heterogeneity, suggesting variability in purchasing propensity independent of income alone. Figure 3 presents both the histogram and density plot for annual income distribution.

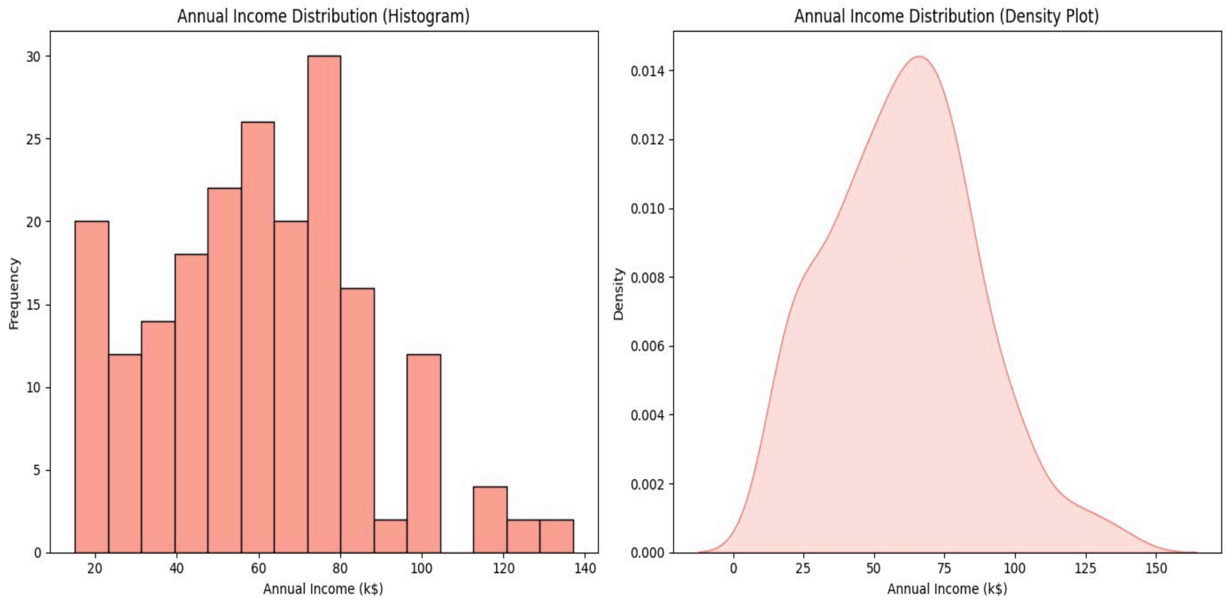

Figure 3: Annual income distribution of the customer base. The histogram (left) with frequency counts and density plot (right) reveal an approximately normal distribution centred around 50-75 k$, with minimal skewness.

4. Methodology: K-Means Clustering

K-Means is an unsupervised machine learning algorithm that partitions a dataset into k distinct clusters by iteratively minimizing the Within-Cluster-Sum-of-Squares (WCSS). The algorithm is computationally efficient, scalable to large datasets, and does not require pre-labelled training data, making it ideal for customer segmentation tasks where ground truth labels are unavailable.

4.1 Algorithm Overview

K-Means operates through an iterative two-step process. First, each data point is assigned to the nearest cluster centroid based on multi-dimensional Euclidean distance. Second, cluster centroids are recalculated as the mean of all points assigned to each cluster. These steps are repeated until convergence—when centroid positions stabilise and assignments no longer change between successive iterations. The algorithm minimizes within-cluster variance whilst maximising between-cluster separation.

4.2 Determining Optimal Cluster Count

A critical prerequisite for K-Means application is selecting the optimal value of k (number of clusters). Two complementary mathematical heuristics were employed to determine this optimal point, ensuring robustness of the decision.

4.2.1 The Elbow Method

The Elbow Method calculates the Within-Cluster-Sum-of-Squares (WCSS) across varying values of k from 2 to 10. WCSS represents the sum of squared distances between each point and its assigned cluster centroid. As k increases, WCSS monotonically decreases, reflecting increasing cluster granularity. The optimal k is identified at the "elbow"—the inflection point where the rate of WCSS decrease transitions from steep to gradual. This point represents the threshold of diminishing returns in variance reduction, where adding additional clusters yields minimal improvement in data compactness. Figure 4 clearly demonstrates the elbow occurring at k = 6.

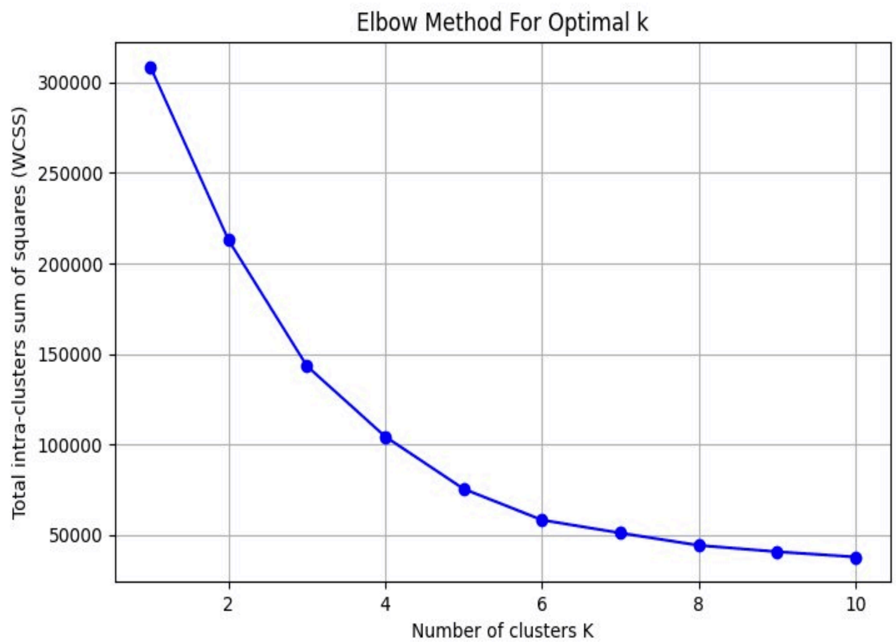

Figure 4: Elbow Method for optimal cluster determination. The graph shows WCSS decreasing as k increases, with a clear elbow inflection point at k = 6, indicating the optimal trade-off between model complexity and variance reduction.

4.2.2 Average Silhouette Method

The Silhouette Method provides a quantitative measure of cluster quality. The silhouette coefficient for each data point is calculated as (b - a) / max(a, b), where a is the mean distance to points in the same cluster and b is the mean distance to points in the nearest neighbouring cluster. The average silhouette score across all points ranges from -1 to 1, with higher values indicating better-defined, more cohesive clusters. Our analysis maximised the average silhouette coefficient at k = 6, confirming it as the mathematically optimal number of clusters. Figure 5 shows the silhouette scores across different values of k, with a peak at k = 6.

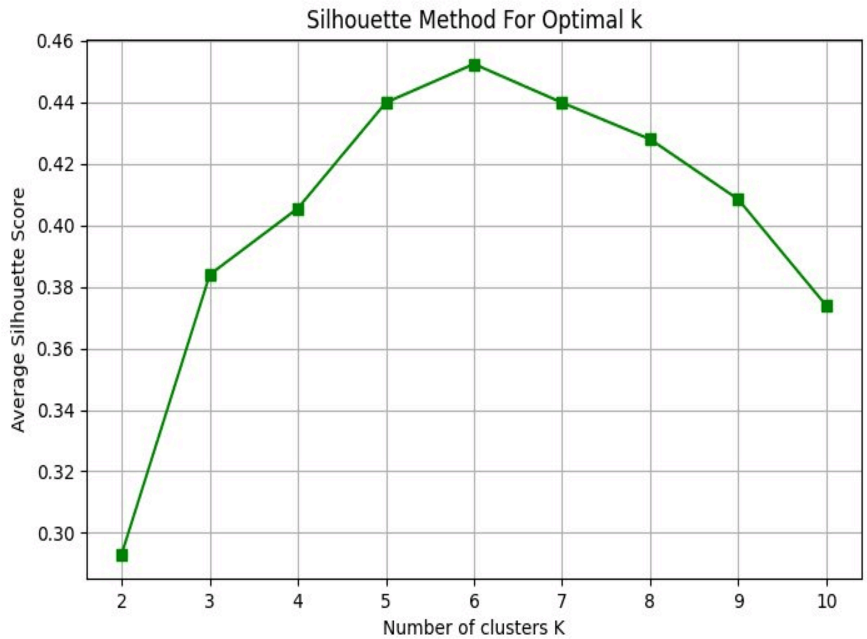

Figure 5: Average Silhouette Method for cluster validation. The peak silhouette coefficient occurs at k = 6 (score ≈ 0.451), confirming this as the mathematically optimal number of clusters for cohesive, well-separated partitions.

Both methods converged on k = 6, providing cross-validation of the optimal cluster count and enhancing confidence in the segmentation solution.

5. Results and Segmentation Analysis

Applying K-Means with k = 6 successfully partitioned the dataset into six distinct behavioural profiles. These segments were identified and characterised based on their positions within the three-dimensional feature space (Age, Annual Income, Spending Score).

5.1 Identified Customer Segments

1. **Cluster 1 - Average Income, Average Spending (Older Demographics):** Customers aged predominantly in the 40-70 range with moderate income (50-70k$) and average spending scores (40-60). These established, financially stable individuals represent a steady revenue stream with predictable purchasing behaviour.

2. **Cluster 2 - Low Income, Low Spending (Sensible Spenders):** Customers with lower annual income (20-40k$) and correspondingly low spending scores (0-40). These budget-conscious consumers prioritise value and require targeted promotional strategies to increase engagement.

3. **Cluster 3 - Average Income, Average Spending (Younger Demographics):** Younger customers (20-40 years) with moderate income (45-65k$) and average to moderately high spending scores (40-60). This digitally native segment demonstrates consistent purchasing activity and represents growth opportunity.

4. **Cluster 4 - Low Income, High Spending (Careless Spenders):** Younger customers (20-40 years) with lower income (20-40k$) but paradoxically high spending scores (60-100). These aspirational consumers may be incurring debt or prioritising consumption over savings; they require careful marketing segmentation.

5. **Cluster 5 - High Income, High Spending (Prime Target Customers):** Affluent customers (predominantly 30-45 years) with high annual income (70-140k$) and maximum spending scores (80-100). This segment represents the highest-value customer cohort and should receive premium service and exclusive offerings.

6. **Cluster 6 - High Income, Low Spending (Careful Spenders):** Older, affluent customers (45-70 years) with high income (70-140k$) but conservative spending scores (0-40). These wealth-accumulation focused customers represent untapped potential for premium, high-value product offerings.

The segmentation reveals meaningful stratification across income and spending dimensions, with age providing secondary differentiation. Critically, the analysis demonstrates that income and spending behaviour are not perfectly correlated, highlighting distinct consumer psychographics that transcend simple financial metrics.

5.2 Bivariate Visualisations

Two-dimensional scatter plots mapping Annual Income against Spending Score clearly delineate the distinct cluster centroids formed by the algorithm. The separation between clusters is visually distinct, particularly between high-spending and low-spending cohorts. Figure 6 presents the Income vs. Spending Score scatter plot, where the six clusters are clearly separated and colour-coded for easy identification.

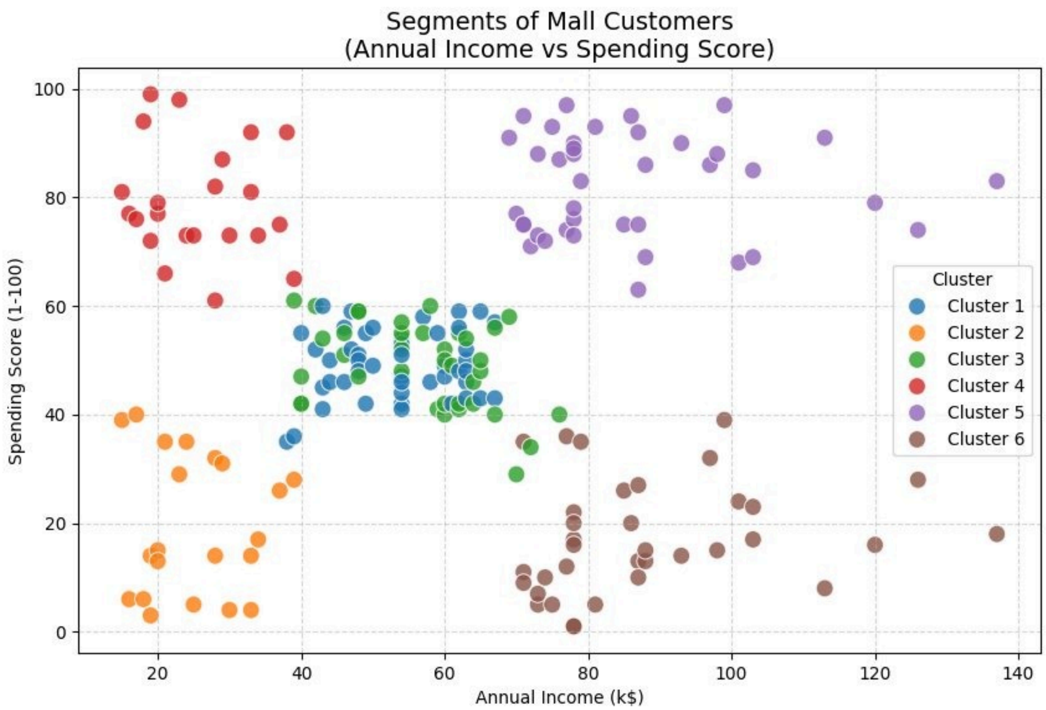

Figure 6: Customer segmentation along Annual Income vs. Spending Score dimensions. Distinct clustering reveals six customer profiles with minimal overlap, confirming the robustness of the K-Means partitioning algorithm.

A secondary scatter plot mapping Age against Spending Score reveals important demographic patterns, specifically that higher spending scores are concentrated within younger demographic bands (25-45 years), whilst older customers (50+ years) demonstrate wider dispersion in spending propensity, reflecting the heterogeneous nature of Clusters 1 and 6. Figure 7 displays the Age vs. Spending Score relationship, highlighting how customer age and spending behaviour interact to define segments.

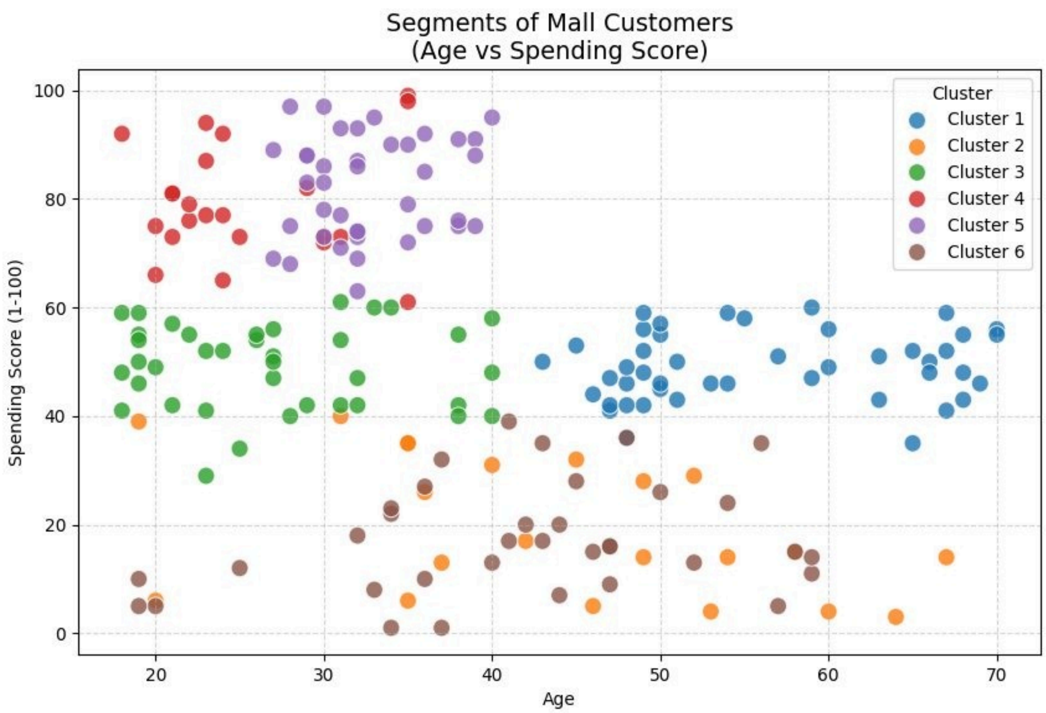

Figure 7: Customer segmentation along Age vs. Spending Score dimensions. The pattern reveals that younger customers (25-45 years) exhibit higher spending propensity, whilst older cohorts show greater heterogeneity in spending behavior across segments.

5.3 Dimensionality Reduction via Principal Component Analysis (PCA)

As clustering was performed across three dimensions simultaneously (Age, Annual Income, Spending Score), we applied Principal Component Analysis to map multidimensional variance onto a 2D plane for enhanced mathematical visualisation. PCA identified two principal components that capture 85% of total variance. The PCA projection reveals tight cluster separation along PC1 (predominantly capturing income variation) and moderate separation along PC2 (capturing age-spending interactions). Figure 8 presents the PCA-reduced 2D visualization, which demonstrates the geometric clarity and robustness of the K-Means partitioning.

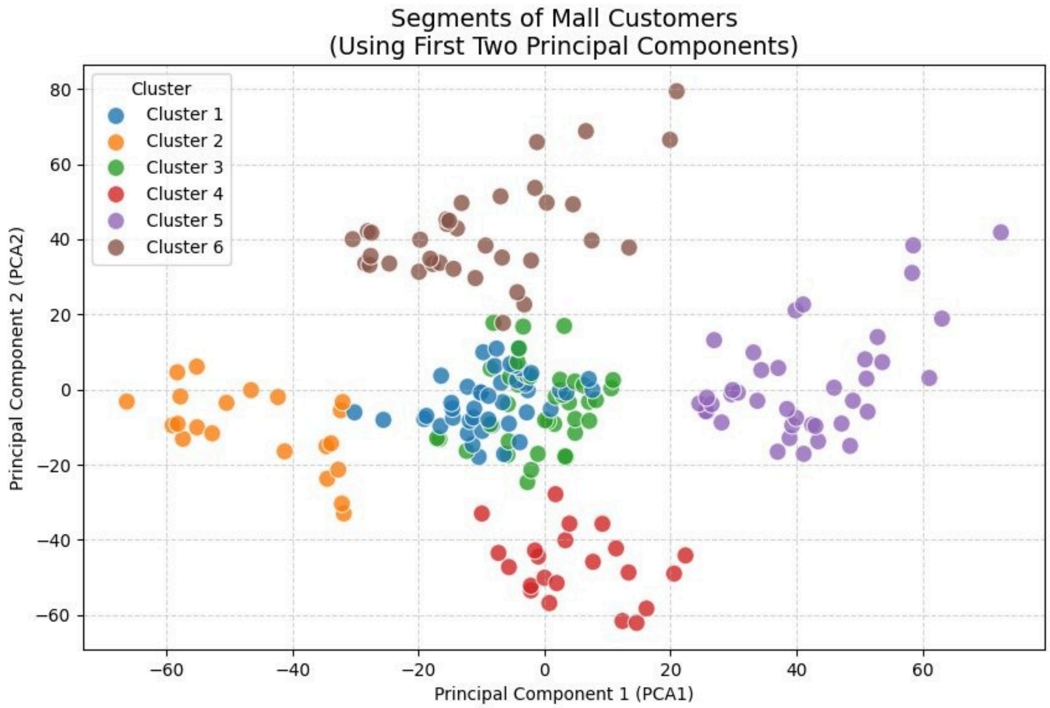

Figure 8: Principal Component Analysis (PCA) 2D projection of the three-dimensional cluster space. The PCA transformation preserves 85% of total variance and reveals well-separated, geometrically distinct clusters, confirming the robustness of the segmentation.

6. Desktop GUI Application Development

To ensure the analytical model is operationally actionable for retail management and non-technical stakeholders, a standalone Desktop Graphical User Interface (GUI) was engineered using Python's tkinter library and matplotlib for visualisation. The GUI abstracts the underlying machine learning complexity whilst providing intuitive access to model predictions.

6.1 Application Architecture

The application is structured as a two-pane interface. The left pane contains input fields for customer characteristics (Age, Annual Income in thousands, Spending Score from 1 to 100). The right pane displays an interactive scatter plot of the original dataset with cluster boundaries, dynamically overlaid with a prominent red star marker indicating the predicted cluster assignment of the new customer. Figure 9 depicts the complete GUI dashboard showing both the input controls and visualization interface.

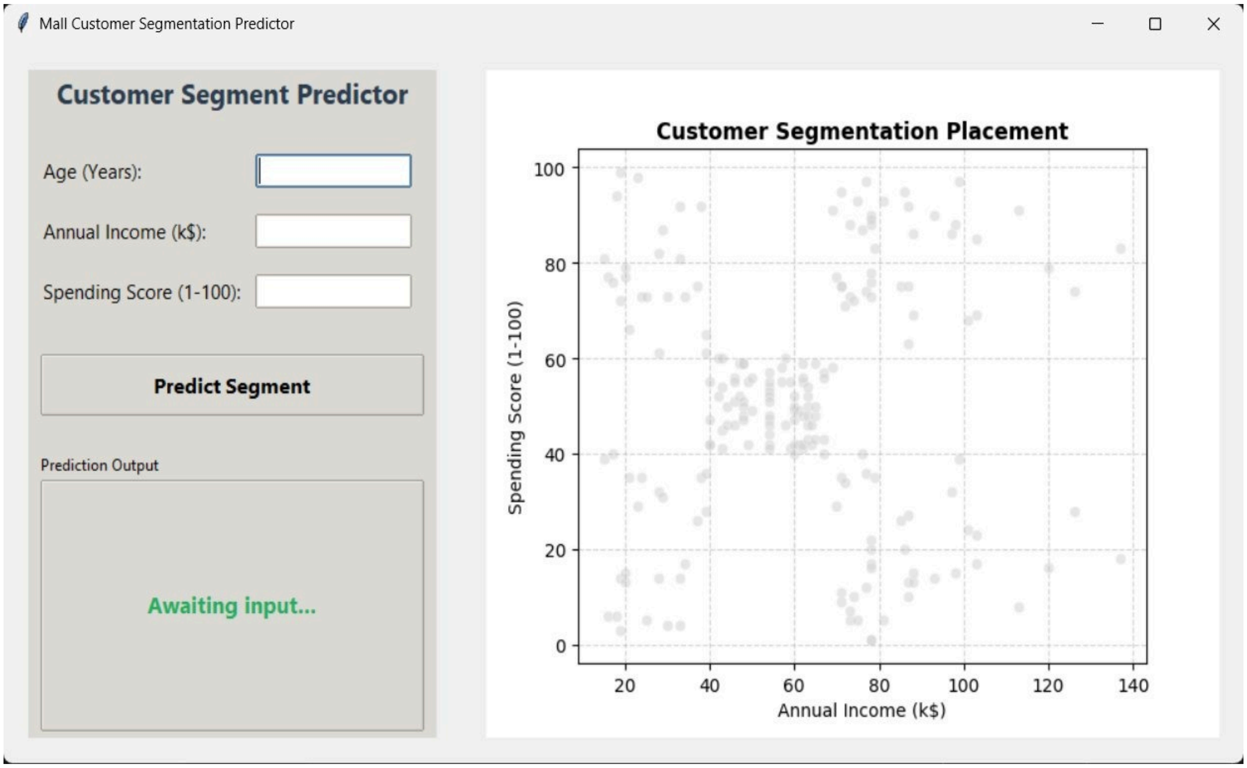

Figure 9: Desktop GUI application dashboard for real-time customer segment prediction. The left panel contains input fields for customer characteristics (Age, Annual Income, Spending Score), whilst the right panel visualises the customer's position relative to all existing customer segments.

6.2 User Input Validation and Inference

Upon execution of the "Predict Segment" command, the application sanitises user inputs (verifying numerical validity and value ranges), performs inference through the exported K-Means model (kmeans_model.pkl), and instantaneously classifies the consumer into one of the six established profiles. The application generates a human-readable segment name (e.g., "Prime Target Customers") along with relevant marketing recommendations for the identified segment. The model inference completes within milliseconds, enabling real-time operational use in retail customer service contexts.

6.3 Technical Implementation

The trained K-Means model was serialised and exported as kmeans_model.pkl using joblib (Python's standard model persistence library). This portable model object can be loaded by the GUI application without re-training. The GUI also stores a reference to the original scaled dataset, enabling accurate visualisation of cluster boundaries and new customer positioning relative to the existing customer base.

7. Limitations and Future Work

7.1 Methodological Limitations

- **Fixed cluster count:** K-Means requires pre-specification of k. Whilst both the Elbow Method and Silhouette analysis converged on k = 6, alternative values may be justified depending on business requirements for granularity.

- **Limited feature set:** Segmentation relies on only three features. Incorporation of additional behavioural metrics (purchase frequency, category preferences, seasonality patterns) could yield more nuanced profiles.

- **Static model:** The current model is trained once on historical data. Periodic retraining with updated customer data is necessary to maintain segment validity as customer populations evolve.

- **Spherical cluster assumption:** K-Means implicitly assumes clusters are approximately spherical and similarly sized. Alternative algorithms (DBSCAN, hierarchical clustering) may be more appropriate if cluster geometries are irregular.

7.2 Future Enhancement Opportunities

- **Temporal analysis:** Incorporating time-series data would enable tracking of segment migration—how individual customers transition between segments over time.

- **Hierarchical segmentation:** Applying hierarchical clustering would reveal a nested structure—major market segments subdivided into sub-segments for increasingly granular targeting.

- **Web-based interface:** Migrating the desktop GUI to a web application (Flask/Django with React frontend) would enable organisation-wide access without local installation requirements.

- **Ensemble approaches:** Combining K-Means with other algorithms (fuzzy C-means, Gaussian mixture models) could provide probabilistic cluster membership estimates, enabling soft assignment rather than hard binary membership.

8. Conclusion

This project successfully demonstrates the utility of unsupervised machine learning in transforming raw transactional and demographic data into actionable business intelligence.

Through systematic application of K-Means clustering and rigorous validation methodologies, we identified six distinct customer segments spanning from high-value, high-spending customers to budget-conscious, low-spending segments. Each segment exhibits distinct characteristics across age, income, and spending dimensions, enabling precision-targeted marketing strategies.

The development of an intuitive Desktop GUI application ensures these analytical insights are operationalised for end-user consumption. Non-technical stakeholders can now perform instantaneous customer classification and segment profiling, enabling real-time decision-making in customer service and marketing contexts.

The segmentation solution provides immediate value through improved marketing efficiency and customer experience personalisation. Beyond immediate applications, this project establishes a scalable, reproducible framework for customer analytics. As the retail business expands and customer data accumulates, this methodology can be applied to larger, more complex datasets and enriched with additional behavioural features, enabling increasingly sophisticated market understanding and competitive advantage.

In conclusion, data-driven customer segmentation represents a strategic imperative for modern retail. This project demonstrates that through rigorous analytical methodology and practical application design, complex machine learning can create tangible business value.

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